The Arc page viewer

A short field guide, in a PDF about itself

页面查看器简明指南

This document was written to exercise a PDF viewer: it has an outline, page labels, internal and external links, a landscape page, English and Chinese text and a few drawings. Its text is original and may be reused freely.

本文档用于测试 PDF 查看器：它包含目录、页码标签、内部与外部链接、横向页面、中英文文字和一些图形。

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Further reading

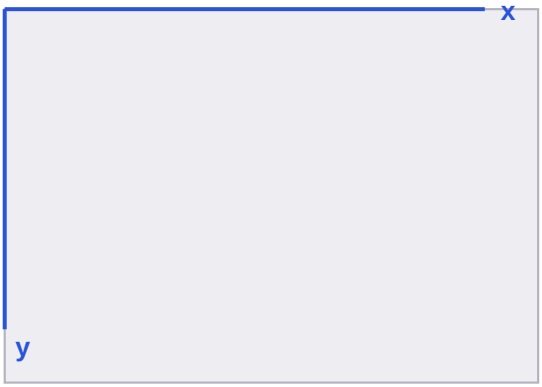
The source of this library: https://github.com/AIDotNet/TKCode.Arc

1. Pages and coordinates

页面与坐标

A page viewer measures everything in page units. One unit at one hundred percent zoom is one device independent pixel, so a page of A4 paper is about 793 by 1123 units whatever the screen does. The origin sits in the top left corner of the page and y grows downward. A PDF file counts from the bottom left in points, and a page may also carry a rotation; the backend converts all of that, so layers and links never have to know.

查看器用页面单位度量一切。百分之百缩放时，一个单位就是一个设备无关像素。原点在页面左上角，纵坐标向下增长。文件里的坐标从左下角开始，并且页面可能带有旋转，这些换算都由后端完成。

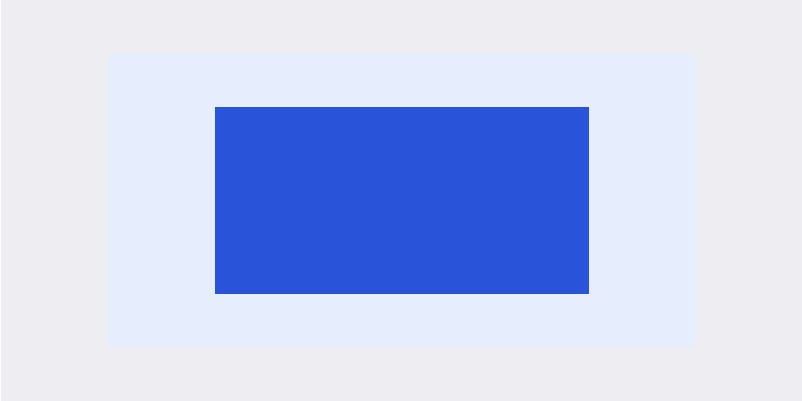


2. Zoom and fit

缩放与适应

Zoom scales the page around an anchor. The wheel and a pinch keep the point under the pointer still, while buttons and keys keep the middle of the viewport still. Fit modes recompute the zoom whenever the viewport changes: fit width makes a line of text as long as the window allows, fit height shows the whole height, and fit page shows the whole sheet. Changing the zoom by hand leaves the fit mode.

缩放以一个锚点为中心。滚轮和捏合让指针下的点保持不动，按钮和键盘让视口中心保持不动。适应模式会在视口变化时重新计算缩放比例。

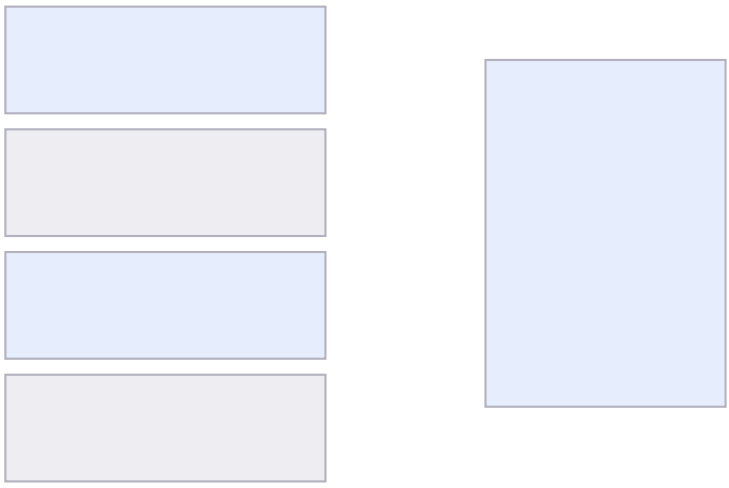


3. Continuous and single page

连续与单页

In continuous layout the pages form one long column and scrolling moves through the whole document. In single page layout one page fills the view and the next one replaces it. Pages of different sizes are centered in their column; this document has a landscape page at the end to show it. Because the presenter draws pages itself, a document of ten thousand pages costs no more visual tree than a document of one.

连续布局把所有页面排成一列，滚动即可浏览全文；单页布局一次只显示一页。不同尺寸的页面会在列内居中，本文档最后一页就是横向页面。

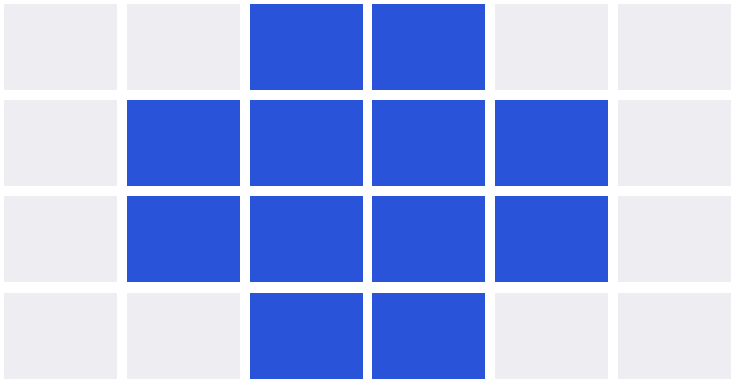


4. Rendering on demand

按需渲染

A page turns into pixels only when it is near the viewport. Requests nearest to the middle of the view go first, and requests for pages that scrolled away are cancelled before they start. While the zoom moves, the existing bitmap is stretched; a sharp one is requested once it settles. Very large zooms draw a capped page plus a detail tile for what is actually visible.

页面只有接近视口时才会变成像素。离视口中心最近的请求优先，滚出视口的请求在开始之前就被取消。缩放进行时直接拉伸已有位图，停止后再请求清晰版本。

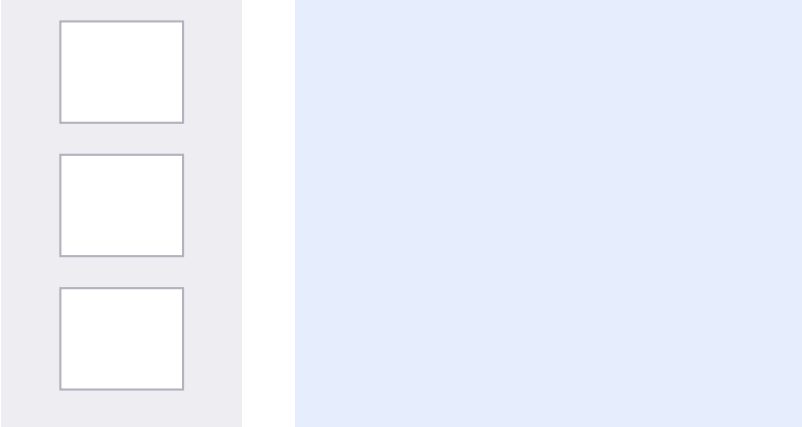


5. Thumbnails and the outline

缩略图与目录

The sidebar has two pages. Thumbnails are small renders that go through the same scheduler and cache as the main view, so a hidden list costs nothing. The outline is the table of contents that the document itself carries; this file has one with two levels and titles in two languages. A document without an outline simply does not show that page of the sidebar.

侧栏有两页。缩略图和主视图共用同一个调度器和缓存，隐藏时不产生任何开销。目录来自文档本身，本文件的目录有两层，标题使用两种语言。

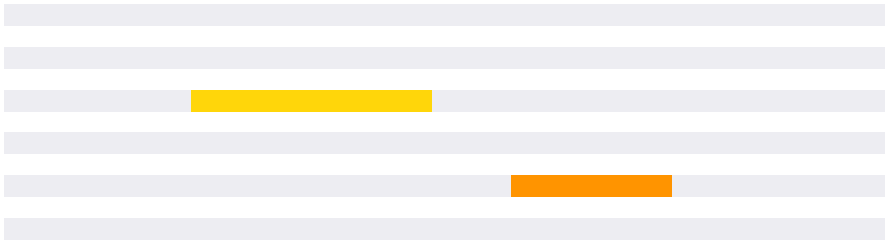


6. Searching text

搜索文字

Search works on plain text with a box for every character, which the backend provides. Matching ignores case unless asked not to, folds compatibility forms such as the ligature in the word office, and joins a word that a hyphen split across two lines. The word lighthouse appears exactly once in this document, on this page, so a test or a person can check that a search finds it.

搜索基于纯文本和每个字符的位置框。匹配默认忽略大小写，并能合并被连字符拆到两行的单词。灯塔这个词在本文档中只出现一次，就在这一页。



7. Selecting and copying

选择与复制

Dragging over text selects it, a double click takes a word and a triple click takes a line. A selection may run across pages and copies in reading order with a line break between pages. The text layer of a page is read only when the pointer first enters that page, so documents with thousands of pages open as quickly as short ones.

在文字上拖动即可选择，双击选词，三击选行。选区可以跨页，复制时按阅读顺序拼接。文字层只在指针第一次进入页面时才读取。

8. Links

链接

A link is either a place in the document or an address outside it. Internal links scroll to their target. External links open only for the http, https and mailto schemes, because a PDF file is untrusted input and a link could lead anywhere else. Addresses that are merely text on the page are detected too, like the one in the box below.

链接要么指向文档内的位置，要么指向外部地址。内部链接滚动到目标；外部链接只接受安全的协议，因为文档是不可信的输入。

This address is only text on the page:

https://www.example.com/arc/pdf-guide

Back to the contents Visit the repository

9. Passwords and untrusted files

密码与不可信文件

A locked document asks for its password in a small form, and asks again when the password is wrong. The viewer keeps the password only for as long as opening takes, and it never reaches a log or an error message. Files can be hostile in other ways: an outline that runs in a circle is cut off, and scripts inside a document never run.

受保护的文档会在表单中询问密码，密码错误时再次询问。密码只在打开期间保留，不会写入日志或错误消息。查看器也会防御循环目录等恶意内容。

Appendix: a wide page 附录：横向页面

This page is landscape, wider than the others, so continuous layout has to center pages of different widths. The table lists the keys a viewer answers to.

Keys	What they do
Arrow keys	Scroll by a small step
Page Up, Page Down	Scroll by a screen, or turn a page in single page layout
Home, End	First page, last page
Ctrl or Cmd with + and -	Zoom in, zoom out
Ctrl or Cmd with 0	Back to one hundred percent
Ctrl or Cmd with F	Open search